Time to Collapse: Institutional Fragility and Norm Evolution in LLM Swarms under Predatory Defection

Carlos Nobuyuki Noda Júnior
Virtual State University of São Paulo

Alan S. Profeta
Federal University of São Carlos

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Abstract

The deployment of autonomous LLM-based agents in shared-resource environments requires governance mechanisms capable of preventing collapse through overexploitation. Although moral reasoning such as universalization reduces the risk of collapse in Common-Pool Resources (CPRs), its resilience to exogenous defection remains underexplored. In this preliminary exploratory study, we investigate the impact of introducing a single greedy agent (newcomer) in Month 4 of a 4-agent commons (qwen2.5:7b) governed by universalization in the GovSim environment (N=40 runs). A landmark analysis from Month 4 indicates that the defector's arrival reduces long-term survival from 87.5% in the control to 33.3% ($p = 0.024$) among runs still alive in Month 4. Our initial findings suggest that collapse is concentrated in the month of arrival (5 of 12 groups). Furthermore, we observe that cooperative swarms spontaneously develop norm tightening (median of 3 t/agent, -0.24 t/month), whereas the perturbation causes systematic normative inflation (+0.68 t/month, $p =$

0.006). These indications suggest that swarm vulnerability stems from the disruption of accumulated normative capital, rather than from a breakdown in compliance with the agreed rules.

1. Introduction

Decentralized resource allocation by autonomous agents based on Large Language Models (LLMs) requires systems to balance the pursuit of individual utility with the preservation of common goods. The Common-Pool Resource (CPR) dilemma represents a critical scenario for testing swarm alignment under physical and dynamic constraints. Although recent approaches indicate that communication combined with moral prompts based on Kantian universalization can prevent the "Tragedy of the Commons" in closed environments, practical deployment exposes these systems to misaligned or exogenous agents. Investigating how cooperative swarms respond to the intrusion of agents pursuing misaligned extraction is essential for designing resilient institutional architectures. In this paper, we present an exploratory analysis of the behavioral and institutional failure modes observed in a swarm of LLM agents (qwen2.5:7b) exposed to the entry of a greedy defector in Month 4 of a 12-month fishing simulation on the GovSim benchmark ($N = 40$ runs).

Our main contributions are:

1. Mapping Emergent Normative Dynamics (Ostrom Paradigm): We identify evidence that cooperative groups spontaneously develop adaptive norm tightening (ratifying quotas of 1 to 4 t/agent, -0.24 t/month), whereas the introduction of the defector triggers systematic normative inflation ($+0.68$ t/month, $p = 0.006$), with no detectable difference in compliance between groups.
2. Exploratory Identification of Fragility to Exogenous Defection: We show preliminarily that, although universalization promotes stability in isolated groups, the arrival of a defector reduces cumulative resource survival from 87.5% to 33.3% (Fisher $p = 0.028$).

2. Related Work

Our research lies at the intersection of governance of LLM-based multi-agent systems, social dilemmas studied in multi-agent reinforcement learning (MARL), and institutional economics. Piatti et al. (2024) introduced GovSim to investigate cooperation among LLM agents in managing common resources, showing that communication and universalization-based reasoning can improve collective sustainability. In MARL, Leibo et al. (2017) formalized sequential social dilemmas to

study how competition for shared resources affects cooperation among agents. Complementarily, Ostrom (1990) provides a theoretical framework for understanding self-governance and the formation of collective rules in common-pool resource systems.

Building on these contributions, we investigate the entry of a greedy agent into a previously cooperative population whose extraction decisions are simultaneous. The landmark analysis allows us to examine group survival from the month of entry onward, although it does not eliminate selection bias or the difference in initial composition between conditions. The results indicate lower survival after the newcomer's entry.

3. Methods

3.1 Environment Setup

Simulations were conducted in the GovSim fishing scenario, with initial biomass fixed at 100 tons, a 100% regeneration rate applied to the stock remaining after the harvest period, and the ecosystem's carrying capacity kept at 100 tons. Depletion of the commons was defined as any time the residual stock fell below 5 tons.
The simulation spans 12 monthly rounds. Each round consists of:

- Natural Communication Phase**:** Discursive exchange among members of the population aimed at negotiating and formalizing the suggested collective harvest quota ($agreed_limit$).
- Concurrent Execution**:** Individual, simultaneous submission of numeric requests of $h_i \in [0, 100]$ tons, subtracted synchronously from the biomass.

3.2 Experimental Design

All incumbent agents run on the qwen2.5:7b model (temperature $T = 0.7$). We analyzed 40 runs distributed across 3 conditions:

- Baseline ($n = 14$): Agents without universalization prompts.
- Universalization Control ($n = 13$): 4 incumbent agents with universalization reasoning prompts
- Newcomer Perturbation ($n = 13$): 4 agents identical to the Control in Months 1–3. In Month 4, a 5th agent ("Luke", the newcomer) joins, programmed with a self-interested extraction goal.

3.3 Statistical Analysis and Norm Structure

The **Landmark Survival Analysis** reset the start of follow-up to Month 4 ($t_0 = 4$) for runs still alive at that mark. To track institutional dynamics, we computed the slope of normative adaptation (β):

$$\beta = \frac{\Delta agreed_limit}{\Delta t} (tons/agent/month)$$

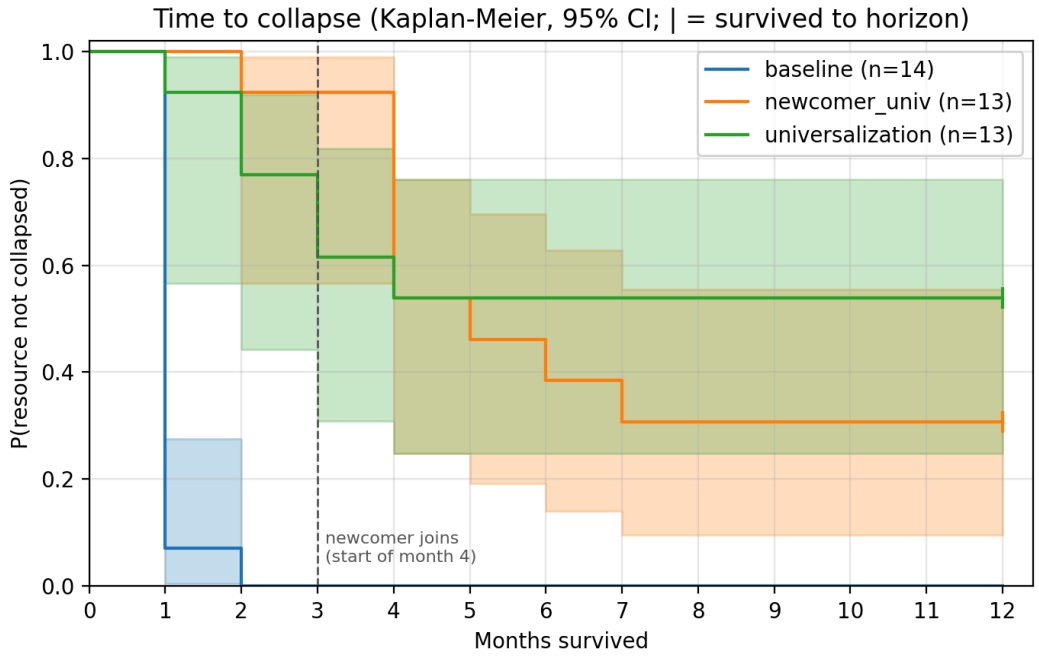
And the Normative Compliance Rate:

$$Compliance\ Rate = \frac{\sum I(h_{incumbent} \leq agreed_limit)}{Total\ Incumbent\ Decisions}$$

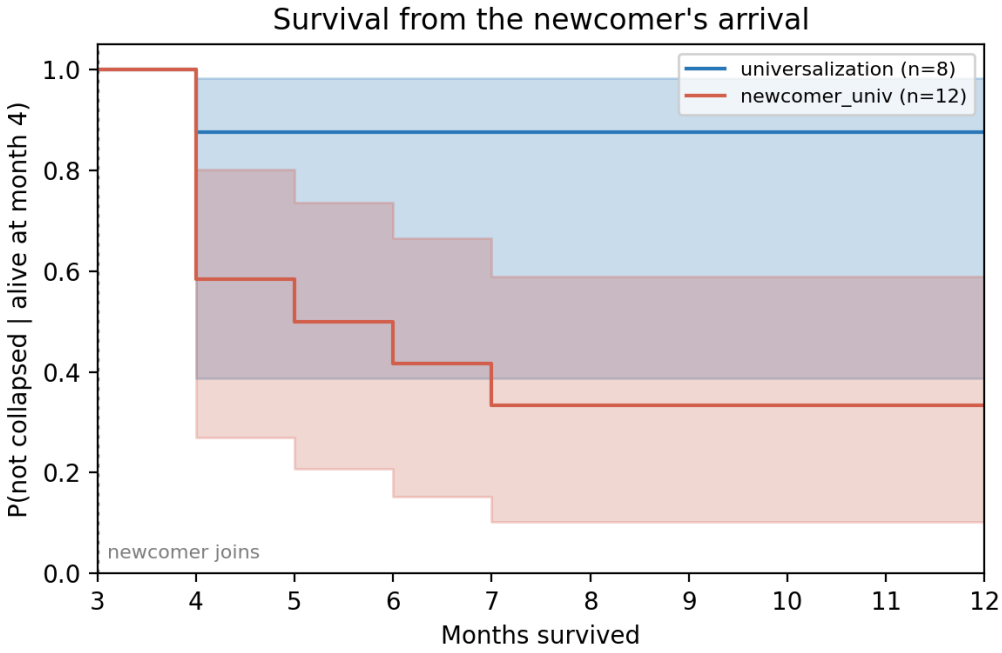
4. Results

4.1 Overall Survival and Month 4 Landmark Analysis

Preliminary survival data indicate the instability of the baseline condition and the initial protection conferred by universalization. The baseline collapsed immediately (median of 1 month; RMST: 1.07 months; $S(12) = 0.00$). Unperturbed universalization maintained 54% survival at Month 12 (RMST: 7.62 months), whereas introducing the newcomer in Month 4 reduced cumulative overall survival to 31% (RMST: 6.77 months; log-rank vs. baseline: $p = 5.4 \times 10^{-6}$ for the control and $p = 5.9 \times 10^{-7}$ for the newcomer).

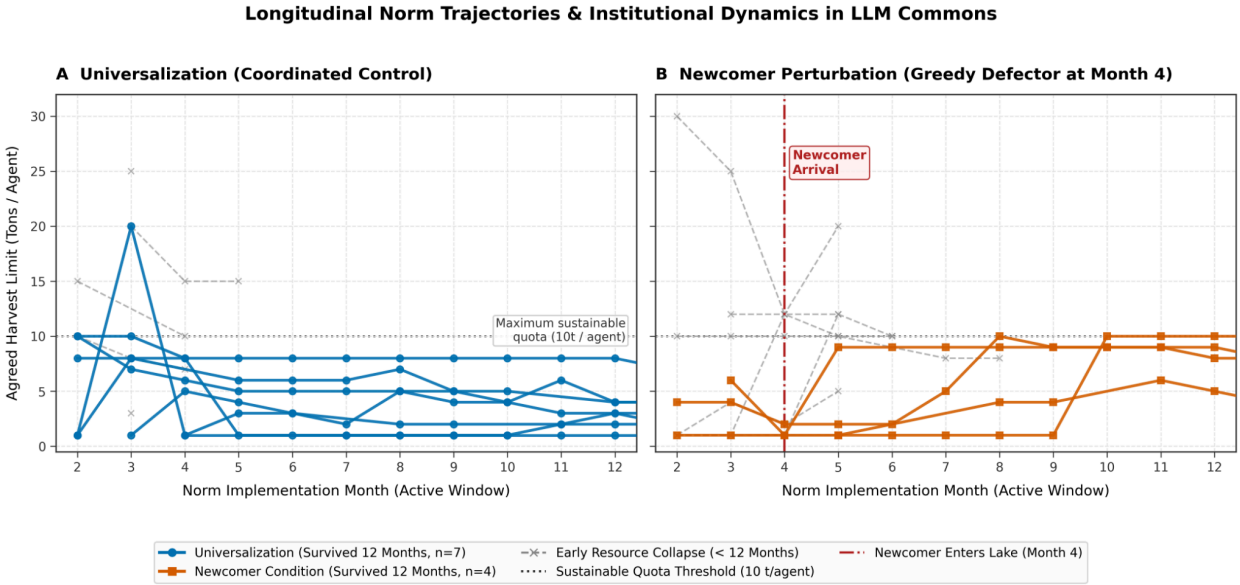


To isolate the effect of the defector's intrusion and mitigate initial biases, we applied a landmark analysis to the groups active in Month 4 ($n = 12$ in Newcomer; $n = 8$ in Control). Of the 12 perturbed groups alive in Month 4, 5 collapsed immediately in that same month and 3 between Months 5 and 12, leaving only 4 groups (33.3%) reaching Month 12, compared with 7 of 8 groups (87.5%) in the Control. Fisher's exact test confirms the reduction in long-term survival ($p = 0.028$, $OR = 0.07$), and the log-rank test over the landmark period validates the divergence in trajectories ($\chi^2 = 5.07$; $p = 0.024$).



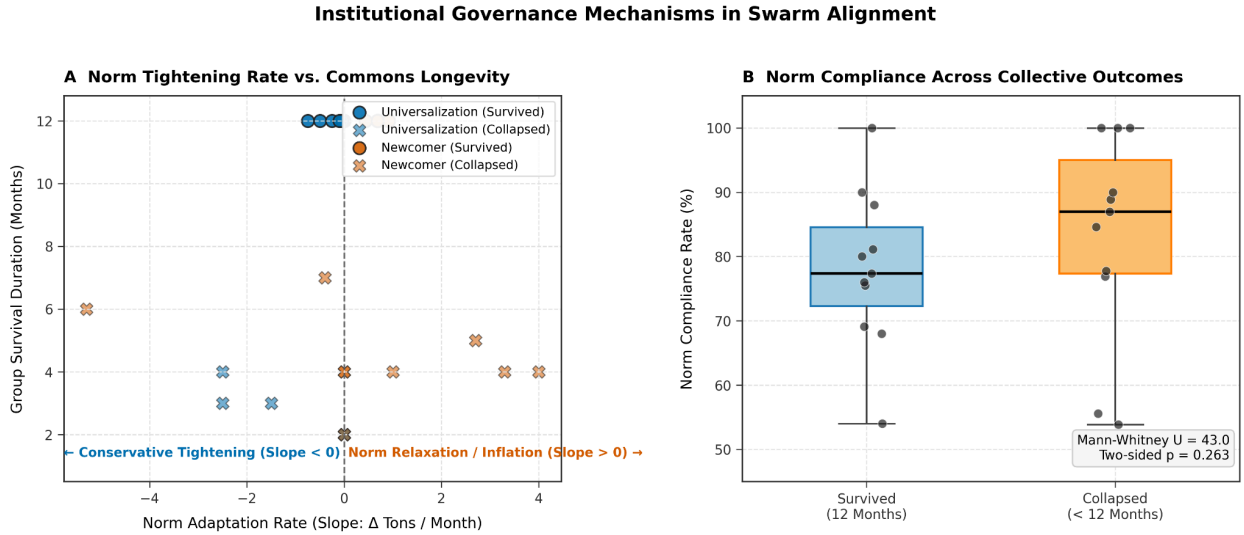
4.2 Emergent Normative Trajectories and Institutional Disruption

Tracking the agreed limits (*agreed_limit*) revealed evidence about the swarm's emergent governance. In the Control, agents exhibited adaptive norm tightening (Elinor Ostrom's Principle 2), spontaneously reducing negotiated quotas to 1 to 4 t/agent. In contrast, the newcomer's intrusion triggered normative inflation in surviving groups, raising the agreed quotas.



As shown in Figure 3, 100% of surviving groups in the Control showed a zero or negative normative slope ($\beta \leq 0$), whereas 100% in the Newcomer treatment exhibited a positive slope ($\beta > 0$; Fisher $p = 0.003$). The mean adaptation rate diverged from -0.238 t/agent/month to

$+\ 0.684$ t/agent/month (Mann-Whitney $p = 0.006$), resulting in significantly higher final limits at Month 12 (median of 8.0 t vs. 3.0 t; $p = 0.024$).



Finally, normative compliance remained statistically equivalent between surviving (78.1%) and collapsed (83.1%; $p = 0.263$) groups. These data preliminarily suggest that collapses do not stem from internal indiscipline against the agreed rules, but rather from the loss of the ecological safety margin caused by quota inflation.

5. Discussion and Limitations

Our exploratory indications suggest that moral alignment prompts (such as universalization) prevent the collapse of the commons in closed ecosystems, but are fragile in the face of exogenous shocks. Notably, we found no difference in compliance between groups. The reversal from adaptive quota tightening to normative inflation signals that the intrusion of a defector erodes accumulated institutional capital, reducing the group's ecological safety margins.

These results should be interpreted with caution due to methodological and scope limitations inherent to exploratory research. The total sample ($N = 40$) and the institutional follow-up subsamples ($n = 7$ in the control and $n = 4$ in the newcomer condition) require larger-scale validation. The study was limited to the qwen2.5:7b model in the GovSim environment, recorded partial drift into Chinese in seven simulations, and assumed strictly synchronous execution with fixed entry in Month 4. Changes to this structure, such as adopting sequential turns with state locks, could significantly alter the group's reaction times. Future work should explore broader model families, heterogeneous populations, and active runtime defensive governance mechanisms.

6. Conclusion

This preliminary investigation indicates that the entry of an exogenous defector agent is associated with reduced long-term survival in LLM swarms: among runs still alive in Month 4, 87.5% of the

control reached Month 12, compared with 33.3% of the newcomer runs (Fisher $p = 0.028$; landmark log-rank $p = 0.024$). In the groups that survived, the defector's arrival was also associated with a reversal in normative dynamics: the adaptive quota tightening observed in the control (−0.24 t/agent/month) gave way to inflation of the agreed quotas (+0.68 t/agent/month; $p = 0.006$), with no detectable difference in rule compliance between surviving and collapsed groups ($p = 0.263$). These results are consistent with the hypothesis that the vulnerability of these swarms is linked to the loss of accumulated normative capital rather than to non-compliance with the agreed rules, but this study does not test that mechanism.

For AI safety, the findings suggest that, in this scenario and with this model, prompt-level moral alignment is not sufficient to maintain the sustainability of multi-agent ecosystems open to exogenous agents. Future work should test whether deterministic governance infrastructure, such as real-time quota monitoring and operational execution locks, preserves resource stability, and investigate the collapse mechanism with larger samples, other models, and a control that includes a fifth cooperative agent.

Code and Data

Include links if applicable. If your project doesn't involve code (e.g., policy analysis) or if there are info-hazard considerations, note that here.

- **Code repository**: https://github.com/NodaNobuyuki/time-to-collapse

References

1. Piatti, G., Jin, Z., Kleiman-Weiner, M., Schölkopf, B., Sachan, M., & Mihalcea, R. (2024). Cooperate or Collapse: Emergence of Sustainable Cooperation in a Society of LLM Agents. Advances in Neural Information Processing Systems, 37. https://arxiv.org/abs/2404.16698
2. Ostrom, E. (1990). Governing the Commons: The Evolution of Institutions for Collective Action. Cambridge University Press. https://doi.org/10.1017/CBO9780511807763
3. Leibo, J. Z., Zambaldi, V., Lanctot, M., Marecki, J., & Graepel, T. (2017). Multi-agent Reinforcement Learning in Sequential Social Dilemmas. Proceedings of the 16th International Conference on Autonomous Agents and MultiAgent Systems (AAMAS), 464–473. https://arxiv.org/abs/1702.03037

Appendix (optional)

Supplementary material such as additional figures, detailed methodology, prompts used, extended results, etc.

LLM Usage Statement

We used language models (LLMs) to assist in developing parts of the agent simulation code and to speed up the statistical validation scripts (including Kaplan-Meier survival analysis and *landmark* analyses). In addition, we used LLM assistance to adapt, synthesize, and refine the wording of the team's original drafts, ensuring a more academic, formal, and concise tone to fit the submission's page limit. All theoretical premises, simulation data, statistical calculations, and conclusions presented were reviewed and validated fully independently by the team.